



Gambali Seshasai Chaitanya

Bachelor of Technology

Electronics and Communication Engineering

Indian Institute of Technology (BHU) Varanasi

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EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech ECE	IIT (BHU) Varanasi	IIT (BHU) Varanasi	8.45	2022-2026
Senior Secondary	St. Francis School, Indirapuram	CBSE	96.8%	2021
Matriculation	St. Francis School, Indirapuram	CBSE	97%	2019

EXPERIENCE

- Google** June 2026 - Present
Silicon Engineer (Emulation) Bengaluru, Karnataka
 - Architecting emulation models and platform enablement pipelines for next-generation TPUs, streamlining pre-silicon verification and firmware bring-up.
- National University of Singapore** January 2026 - June 2026
Visiting Student Researcher (with Prof. Flavien Solt) Singapore
 - Implemented a pre-silicon framework to detect microarchitectural transient execution vulnerabilities (e.g., Spectre-style side channels) on RISC-V cores. Designed instrumentation free taint-tracking logic to trace transient leaks
- University of California, Berkeley** September 2025 - January 2026
Visiting Student Researcher (with Prof. Christopher Fletcher) California, USA
 - Ported Cell-IFT instrumented RISC-V cores onto Xilinx Alveo U250 FPGAs to accelerate microarchitectural side-channel leakage analysis using the MileSan dynamic framework.
- Google Summer of Code, 2025** June 2025 - September 2025
Software Development Intern Remote
 - Extended **OpenROAD's** physical design flow to support rectilinear floorplans, enabling non-rectangular floorplaning across physical layout steps.
 - Refactored C++ backend geometry logic, updated SWIG interfaces, and exposed user-facing Tcl commands to support polygonal boundary specifications.
- Jaguar Land Rover** May 2025 - July 2025
Embedded Systems Intern Bengaluru, Karnataka
 - Built a secure two-stage bootloader with Root of Trust verification and AES decryption on ARM Cortex-M7 microcontrollers to protect sensitive firmware IP.
 - Optimized bootloader footprint from 8 KB to 3 KB by tuning linker scripts and section placement for resource-constrained automotive microcontrollers
- RISC-V International @ Linux Foundation Mentorship 2025** March 2025 - June 2025
Project Mentee, Spring 2025 term, Linux Foundation Mentorship Program Remote
 - Developed a **Hardware Abstraction Layer Transitional Library for Software to Hardware Posits For RISC-V**
 - Delivered an upgraded version of SoftPosit-cpp library with native support for 8-bit, 16-bit, and 32-bit Posit math functions, comprehensive unit tests, and performance benchmarks.
 - Built CI/CD pipelines and packaging automation using GitHub Actions, along with a Python wrapper for the library.
- Indian Institute of Technology Bombay** May 2024 - July 2024
Summer Research Intern (with Prof. Sandip Mondal) Bombay, India
 - Developed Embedded C-based applications using Vivado and Xilinx SDK to run on a Digilent Zybo FPGA, creating an **FPGA-based controller** to perform Read, Write and Erase Operations on 2D and 3D SLC and MLC raw **NAND Flash** Chips.
 - Used bit-banging to precisely control the onboard GPIO pins to interface with the Raw NAND using the ONFI Consortium Protocol.
- Google Summer of Code, 2024** May 2024 - August 2024
Software Development Intern Remote
 - Contributed to **Apertium**, an open source machine translation platform. Developed command-line utilities to construct bidirectional dictionaries for a language pair, given a pair of parallel corpora - i.e., the same content in two different languages.

- Worked on creating an ABC script to improve synthesis options in Yosys for creating a balanced synthesis plan, balancing between Speed and Area optimized synthesis.

JOURNALS

- Gambali Seshasai Chaitanya, Aditya D. Arkalgud, Shubham Pande, Ankit Arora, "*An Energy-Efficient, CMOS-Compatible Physical Reservoir Node with Post-Fabrication Tunable Decay Dynamics*", Neuromorphic Computing and Engineering 5.4 (2025): 044010.  

CONFERENCES AND PUBLICATIONS

- Gambali Seshasai Chaitanya, Aditya D. Arkalgud, Sourav De, Ankit Arora, "*Adaptive FeFET-Based LIF Neuron for Regression Tasks on Spiking Neural Networks*", 7th International Conference on Emerging Electronics (ICEE 2025, Oral Presentation)  
- João Pedro Cabral Miranda, Gambali Seshasai Chaitanya, Inbasekaran Perumal, Joshua V. Gyllinsky and Kurt Keville, "*Evaluating Numerical Formats for HPC Using Fractal Benchmarks*", Conference for Next Generation Arithmetic (CoNGA'25, Poster Presentation) 
- Gambali Seshasai Chaitanya, Ankit Arora, "*A Non-Volatile ReRAM and Subthreshold-FET-Based Kernel for Energy-Efficient Temporal Signal Processing Using Reservoir Computing*", 23rd Non Volatile Memory Technology Symposium (NVMTS 2025, Poster Presentation)  
- Gambali Seshasai Chaitanya, Arin Weling, Sandip Mondal, "*Stress Induced Failure Mechanism in NAND Flash Memory Devices*", Electronic Materials Conference 2025 (Poster Presentation) 
- Arin Weling, Gambali Seshasai Chaitanya, Sandip Mondal, "*Thermal-Induced Threshold Voltage Distribution Changes in 3D NAND Flash Cells: Experimental Characterization*", Electronic Materials Conference 2025 (Oral Presentation) 
- Asifa Khatun, Gambali Seshasai Chaitanya, Sandip Mondal, "*Page and Block-wise Program Disturb Operation in NAND Flash Memory*", Electronic Materials Conference 2025 (Poster Presentation) 
- Gambali Seshasai Chaitanya, Shubham Pande, Ankit Arora, "*ReRAM-based Crossbar Compatible Equality Checker for Neuromorphic Applications*", 8th Students' Conference on Engineering & Systems (SCES-2024, Oral Presentation)  
- Gambali Seshasai Chaitanya, Vaseekaran Elangovan, "*ReRAM Crossbar Compatible Efficient Equality Checker and Half Adder Implementation*", XXII International Workshop on Physics of Semiconductor Devices (Oral Presentation)  

PROJECTS

• Design and Tapeout of a 64 Point FFT Module

Feb 2026 - April 2026

- Architected a 64-point FFT module in Verilog, tailored for low-area utilization.
- This project also marks the first successful tapeout from IIT (BHU) Varanasi
- Successfully taped out the design on SkyWater 130nm via the TinyTapeout MPW shuttle, driving the automated OpenLane ASIC flow from RTL to verified GDSII layout.

ACHIEVEMENTS

- 3rd prize winner in the National Level Digital Design Hackathon conducted by the Ministry of Electronics & Information Technology of India (MeitY) under the Chips to Startup (C2S) Program.
- International Finalist in Google Code-In 2019, out of 14,798 participants from 117 countries
- Winner of the 'Startup with German' Scholarship, awarded by Goethe Institute. Felicitated by **Angela Merkel**
- Winner of Times Scholar 2019, Felicitated by the then Vice President of India, **Shri Venkiah Naidu**
- National Finalist in Hack AI, an AI/ML based Hackathon, Techfest, IIT Bombay
- 2nd in FPGA-based round in I-Chip, a Verilog-based event in Udyam'24, EES IIT (BHU) Varanasi
- Finalist in Digisim, a Digital Electronics Event, Udyam'24, EES IIT (BHU) Varanasi